

Qatar-1b

R-1475

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_190523 · created 2026-08-01T12:18:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458626.8467 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.056
Transit depth (Rp/Rs)^2	3.07 +/- 1.98 [%]
Orbital Inclination (inc)	82.81 +/- 2.88
Airmass coefficient 1 (a1)	0.84 +/- 0.64
Airmass coefficient 2 (a2)	0.24 +/- 0.82
Scatter in the residuals of the lightcurve fit is	91.96 %
Best Comparison Star	None
Optimal Aperture	1.06
Optimal Annulus	4.58
Transit Duration (day)	0.056 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.056 vs 0.1463 (deviation 0.5σ)
Duration fit vs published	0.056 d vs 0.0692 d (deviation 0.52σ)
Mid-time O–C	3.4 min
Depth SNR	3.1
Residual scatter	91.96 %

Light curve

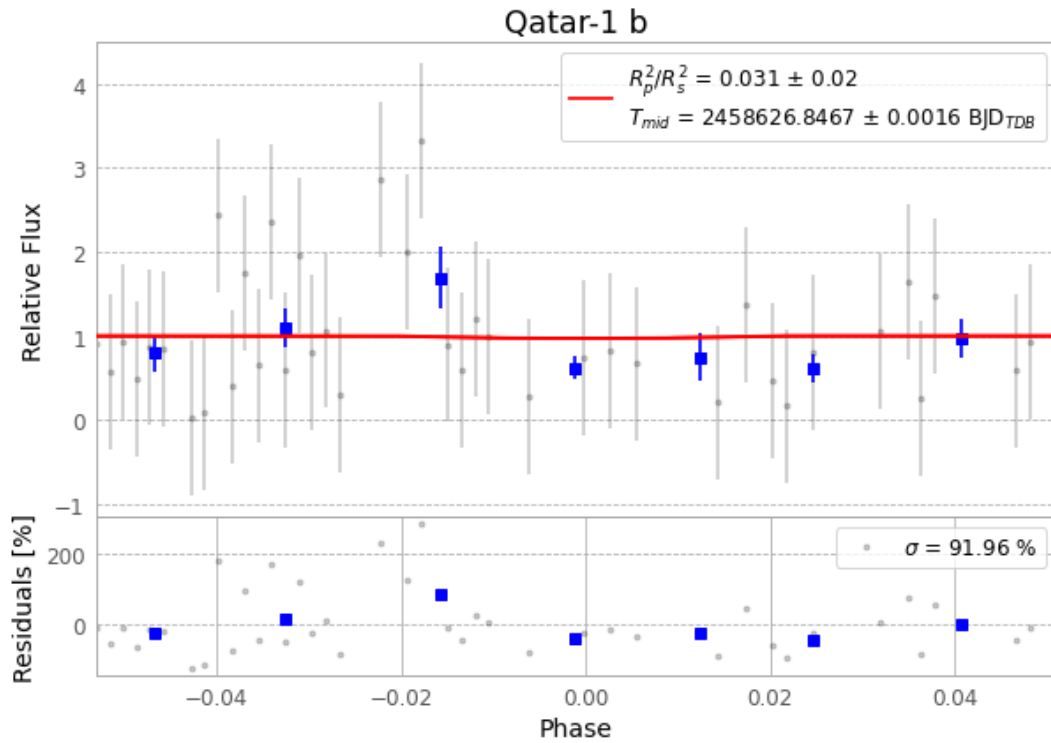


Figure 1 — FinalLightCurve_Qatar-1 b_2019-05-22.png (SHA-256 5dc32564103f1656...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 371], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01618cdd5a8b6010...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), Qatar-1190523063012.FITS → Qatar-1190523100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190523.log (a377b7dbca4c...), FinalLightCurve_Qatar-1 b_2019-05-22.png (5dc32564103f...), FinalLightCurve_Qatar-1 b_2019-05-22.csv (1663307f1e45...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:18Z.